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Chemistry	Paper No.6: PHYSICAL CHEMISTRY-II (Statistical Thermodynamics, Chemical Dynamics, Electrochemistry and Macromolecules)
	Module No. 31: Polymers and their Classification

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Chemistry

Paper No.6: PHYSICAL CHEMISTRY-II (Statistical Thermodynamics, Chemical Dynamics, Electrochemistry and Macromolecules)

Module No. 31: Polymers and their Classification

1. Learning Outcomes

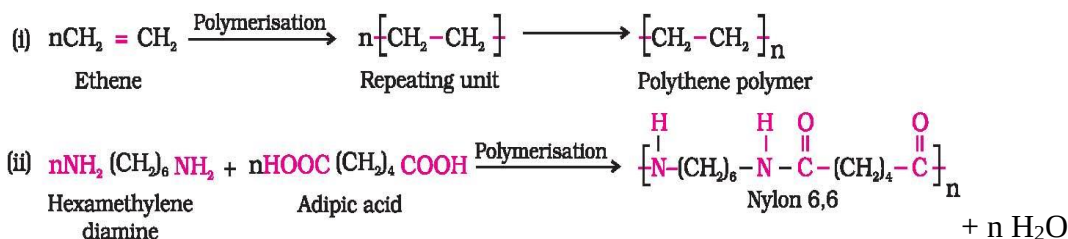
After studying this module, you shall be able to

- Know what are macromolecules
- Learn the various classifications of polymers.
- Understand what is polymerization

2. Introduction

Macromolecules are large molecules, mostly built by joining smaller molecular subunits, called “monomers”. The word **macromolecule** (Greek macros, long, large) was introduced by the German chemist Herman Staudinger (1881-1965) to refer to large molecule of molecular mass of more than 10000 amu. These molecules are also called **polymers**, where “poly” means many and “mer” means structural subunits. In other words, polymers are large sized molecules having high molecular weight, built by joining together numerous small sized molecules, also known as monomers. The reaction in which monomers (subunits) combine to form a polymer (macromolecule) is known as polymerization. Polymerization is a type of chemical reaction in which two or more molecules combine together with or without the evolution of heat or elimination of water to form a molecule of high molecular weight. Thus the starting material is called monomer and the resulting product is called polymer.

A polymer in its elementary form can be regarded as comprising of molecules of closely related composition of molecular mass at least 2000 amu, although in many cases characteristic properties do not become obvious until the average molecular mass is about 5000. There is virtually no upper limit of the range of molecular mass of polymers, since giant three-dimensional networks may produce cross-linked polymers of molecular mass ranging in millions.



Polymers combine the characteristics of both solids and liquids as it exhibits elasticity like solids and viscous behavior like liquids. Hence polymers in solutions have unique viscoelastic, diffusion and colligative properties.

3. Classification of Polymers

Polymer is a general name of a large number of materials with high molecular mass. Polymers vary in the number and types of atoms present in their molecule, which give rise to different geometric structures, physio-chemical properties, mechanical behavior and thermal characteristics. Based on these properties, polymers may be classified in different groups, as given in Table 3.1.

Table 3.1: Classification of Polymers

Basis of Classification	Polymer Type
Origin	Natural, Semi synthetic, Synthetic
Thermal Response	Thermoplastic, Thermosetting
Mode of Formation	Addition, Condensation
Geometric Structure	Linear, Branched, Cross-linked
Applications and Physical Properties	Rubber, Plastic, Fibers
Tacticity	Isotactic, Syndiotactic, Atactic
Crystallinity	Non crystalline (or amorphous), Semi-crystalline, Crystalline

Polarity	Polar, Non polar
Chain	Hetero, Homo-chain

3.1 Origin

Based on their occurrence (or lack of) in nature, polymers can be classified into three types:

A. Natural Polymers: Polymers, which are found to exist in nature, are called natural polymers. Examples of these kinds of polymers include natural rubber, natural silk, wool, DNA, cellulose, starch and proteins.

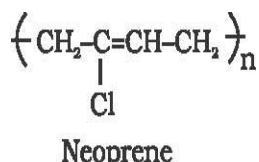
B. Semi synthetic Polymers: These polymers can be obtained by chemical modification of natural polymers, such as hydrogenated natural rubber, cellulosic, cellulose nitrate, methyl cellulose, etc.

C. Synthetic Polymers: Polymers that are chemically produced in the laboratory are known as synthetic polymers or man-made polymers. Common examples of such polymers include polyvinyl alcohol, polyethylene, polystyrene, polysulfone, etc.

3.2 Thermal Response

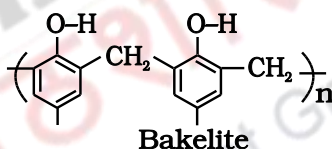
On the basis of temperature variation, polymers can be differentiated into two groups:

A. Thermoplastic polymers: Polymers which get softened or plasticized repeatedly on application of thermal energy, without considerable change in properties if treated with certain precautions are known as thermoplastic polymers. For example – nylons, polyolefins, polyethers, linear polyesters and PVC, sealing wax etc



B. Thermosetting polymers:

Polymers which undergo certain variations on heating and get converted into an infusible mass are known as thermosetting polymers. The 'curing' or 'setting' (conversion) process includes chemical reactions giving rise to further growth and cross linking of the polymer chains/molecules and producing big molecules. For example, phenolic resins, diene rubbers, urea epoxy resins, etc.



3.3 Mode of Formation

Based on the reaction involved in their formation, polymers can be categorized as:

A. Addition polymers: Polymers that are formed from olefinic, diolefinic, vinyl and related unsaturated monomers. These polymers are formed by the addition of monomeric subunits to each other in rapid succession without the elimination of any molecule. This reaction occurs by a chain mechanism and is known as addition polymerization.

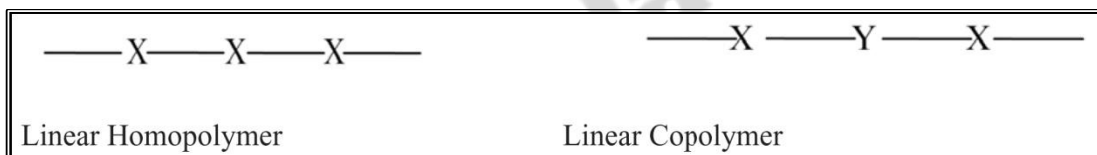
Example: polyethylene, polystyrene, polypropylene, etc.



B. Condensation polymers: Polymers that are formed from the intermolecular reactions between bifunctional or polyfunctional monomer molecules having reactive functional groups such as $-\text{OH}$, $-\text{COOH}$, $-\text{NH}_2$, $-\text{NCO}$, etc. This reaction generally occurs with the elimination of a small molecule as by product, usually H_2O .

On the basis of structure of the polymeric chain so formed, these polymers can be further sub-categorized into three types:

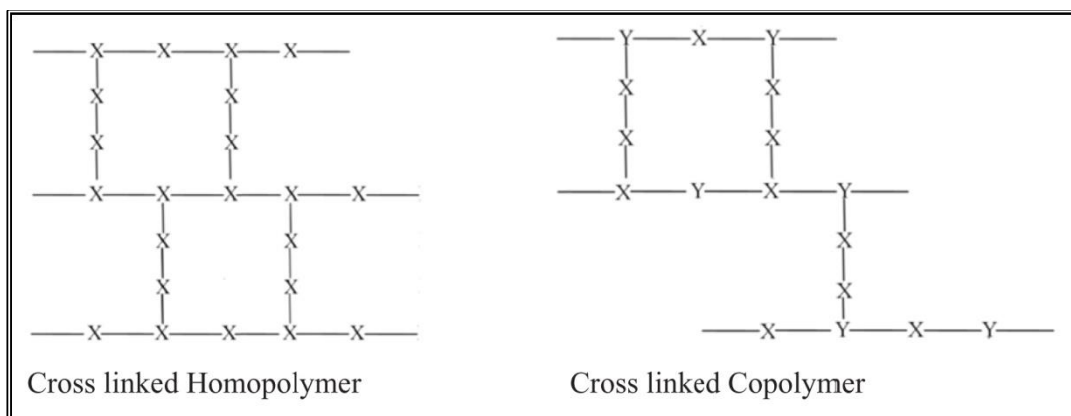
A. Linear Polymers: If the constituent monomers are connected or joined in a linear fashion, then polymer is said to be a linear polymer.



B. Branched Polymers: When the monomer units are joined in a branched manner, then the resultant polymer is known as a branched polymer.



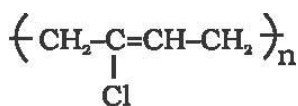
C. Cross linked Polymers: A polymer is said to be cross linked, if different polymer chains are inter-linked or joined in certain places, as following



3.5 Application and Physical Properties

Depending on its ultimate use and form, a polymer can be differentiated as:

A. Rubber (Elastomers): These comprise of rubber-like solids with elastic characteristics. In these polymers, the polymeric chains are joined together by weak intermolecular forces which allow the polymers to be stretched. A few 'crosslinks' are incorporated in between the chains, which allow the polymer to regain its original position after the force is released as in vulcanised rubber. Rubber is a high molecular weight polymer with long flexible chains and weak intermolecular forces. They exhibit tensile strength in the range of 300-30000 psi and elongation at break in the range of 300-1000%. For example natural and synthetic rubber.

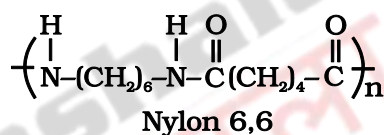


Neoprene

B. Plastics: Plastics are comparatively strong substances with large molecular weight that can be molded with (or without) the application of heat. These are usually stronger

than rubbers. They exhibit tensile strength ranging between 4000-15000 psi and elongation at break ranging usually from 200 to 200% or even higher. The examples of plastics are polyethylene, polypropylene, PVC, polystyrene, etc.

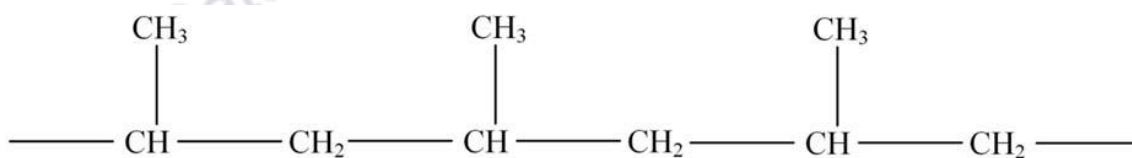
C. Fibers: Fibers are long- chain polymers characterized by crystalline regions resulting mainly from secondary forces. They have a much lower elasticity than plastics and elastomers. They have high tensile strength ranging between 20,000-150,000 psi, are light weight and possess moisture absorption properties.



3.6 Tacticity

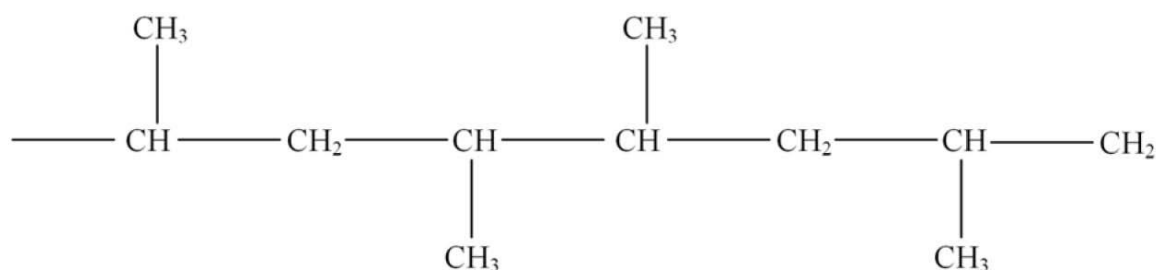
It may be defined as the geometric arrangement (orientation) of the distinct group of the monomer unit with respect to the main chain (backbone) of the polymers. On the basis of structure, polymers are differentiated into three groups.

A. Isotactic polymer: Polymers in which the characteristic and distinctive groups are arranged on the same side of the main chain.



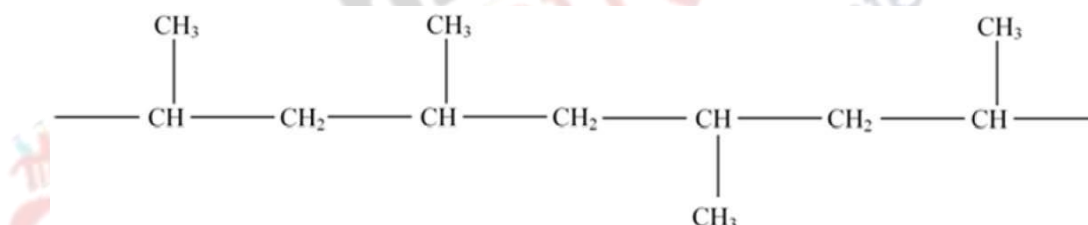
Isotactic Polypropene

B. Syndiotactic polymer: If the characteristic groups are arranged in an alternate fashion along the main chain, then the polymer is said to be syndiotactic polymer.



Syndiotatic Polypropene

C. Atactic polymer:- If the characteristic groups are arranged randomly in an irregular fashion around the main chain, then the polymer is said to be atactic. Such polymers have more strength, effectiveness and elasticity.



Atactic Polypropene

4. Polymerization

Polymers may be categorized in different ways and a convenient way is with reference to the manner of their preparation. The linking of numerous small molecules - termed as monomers- with each other to form a macromolecule (or polymer molecule) through chemical reactions is known as polymerization. It can be defined as the fundamental process by which low molecular weight compounds are converted into high molecular weight compounds. In addition to the structural and compositional differences

between polymers, Flory stressed the characteristic difference in the mechanism by which polymer molecules are built. Although Flory continued to make use of the terms “addition polymerization” (polymerization that occurs by repeated addition process) and “condensation polymerization” (polymerization that occurs by repeated condensation process along with the elimination of small molecules) in his discussion of the polymerization mechanism. The current terminology categorizes polymerization into step growth polymerization and chain growth polymerization.

The degree of polymerization is related to the molecular mass (M) of the polymer and is given by the equation,

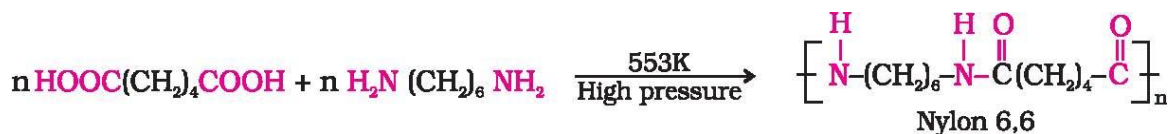
$$D_p = \frac{M}{m}$$

where, m is the mass of the unit monomer.

There are two broad types of polymerization reactions, *i.e.*, the addition or chain growth polymerization and condensation or step growth polymerization.

4.1 Condensation Polymerization or Step-growth Polymerization

Condensation polymerization involves a chemical reaction in which a polymer is formed with the elimination of a small molecule (with lower molecular weight) as a byproduct. The byproduct eliminated is called as condensate. The reaction can occur between two similar or different monomers. In such reactions, each step yields a product which is again a characteristic, bi-functional species. These are independent of each other and so, the chain of condensation goes on. An example of this kind of polymerization – also known as step growth polymerization – includes the formation of terylene (or dacron) by the interaction of ethylene glycol and terephthalic acid.



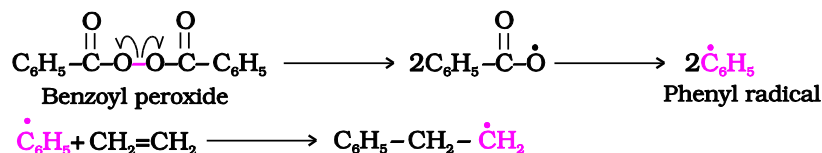
4.2 Addition Polymerization or Chain Polymerization

In addition polymerization, two or more molecules of monomers are linked together to produce a polymer. In this polymerization, there is no elimination of any molecule. It is a chain reaction and no byproduct is released. It is obtained by linking together the monomer molecules through a chain reaction to give a polymer whose molecular weight is exactly an integral multiple of that of the constituent monomers as in the case of polyethylene obtained by polymerization of ethylene. Only a single kind of monomer is involved in this addition polymerization reaction. Therefore polyethylene is a homopolymer, which contains exactly same monomer units. Addition polymerization reaction is generally induced by light, heat or a catalyst for opening the double bond of the monomer and creating the reactive sites

A variety of alkenes, alka-dienes and their derivatives are polymerized in the presence of a free radical generating initiator (catalyst) like acetyl peroxide, benzoyl peroxide, tert-butyl peroxide, etc. For example, the polymerization of ethene to polythene requires heating or, exposing to light – a mixture of ethene with a little amount of benzoyl peroxide initiator. The process begins with the addition of phenyl free radical produced by the peroxide to the ethene double bond thus giving rise to a new and larger free radical. This step – which initiates the chain reaction – is called the **chain initiating step**. As this newly generated free radical reacts with another molecule of ethene, it leads to the formation of another bigger sized radical. The repetition of this sequence of reactions, with the generation of new and bigger free radicals, carries the chain forward and this step is termed as **chain propagating step**. Ultimately, in the **chain terminating step**, the

product free radical reacts with another free radical to form the polymerized product. The sequence of steps may be depicted as follows:

Chain initiation steps

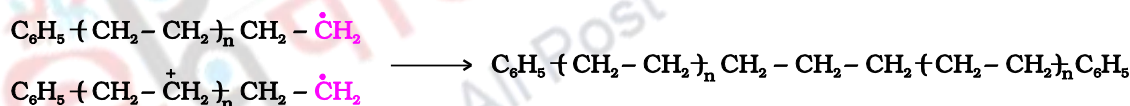


Chain propagating step



Chain terminating step

For termination of the long chain, these free radicals can combine in different ways to form polythene. One mode of termination of chain is shown as under:



5. Summary

Let us summarize what we have learnt:

- Macromolecules are large molecules, mostly built by joining of smaller molecular subunits, called monomers”.
- The reaction by which polymer is formed by combination of monomers is known as polymerization.
- A polymer in its elementary form can be regarded as comprising of molecules of closely related composition of molecular weight at least 2000, although in many cases characteristic properties do not become obvious until the average molecular weight is about 5000.
- Polymer is a general name of a large number of materials with high molecular weight.
- Polymers vary in the number and types of atoms present in their molecule, which give rise to different chemical structures, physical properties, mechanical behavior and thermal characteristics.
- On the basis of heating, polymers can be differentiated into two groups: thermoplastics polymer and thermosetting.
- In addition polymerization, two or more molecules of monomers are linked together to produce a polymer. In this polymerization, there is no elimination of any molecule